# Machine Learning–Based Safety Compliance Classification Using Camera Vision

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## INTRODUCTION

### 1. Project Introduction

This project classifies worker images into Helmet and No Helmet to support workplace-safety monitoring.

Manual monitoring has many limitations:

- Requires a large amount of human resources
- Difficult to monitor continuously
- Easily misses violations
- Low efficiency and high cost

Camera images flag potential helmet violations for review.

### 2. Problem definition

**Task Type**

Binary image classification

**Classes**

- Helmet
- No Helmet

**Input**

Image of worker



**Output**

Predicted class:

Helmet



No Helmet






## Model & Training

MobileNetV2, implemented in TensorFlow/Keras, was adapted by transfer learning using preprocessed and augmented images.

```python
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras import layers, models

base_model = MobileNetV2(
    input_shape=IMG_SHAPE,
    include_top=False,
    weights='imagenet'
)
```

```python
base_model.trainable = False

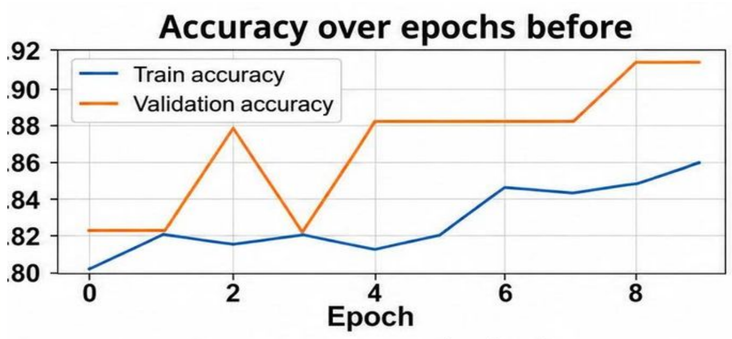
model = models.Sequential([
    base_model,
    layers.GlobalAveragePooling2D(),
    layers.Dropout(0.2),
    layers.Dense(1, activation='sigmoid')
])
```

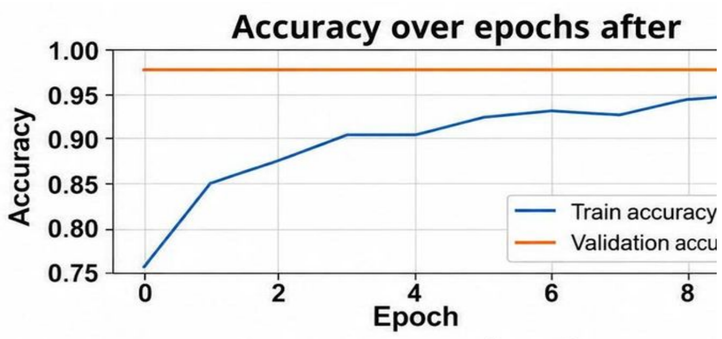
### Optimization and Deployment of the Model

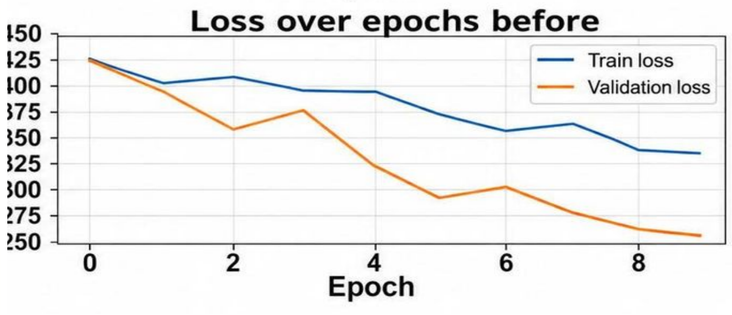
```python
l.compile(
optimizer=
tf.keras.optimizers.Adam(learning_rate=0.000
loss='binary_crossentropy',
metrics=['accuracy']
```

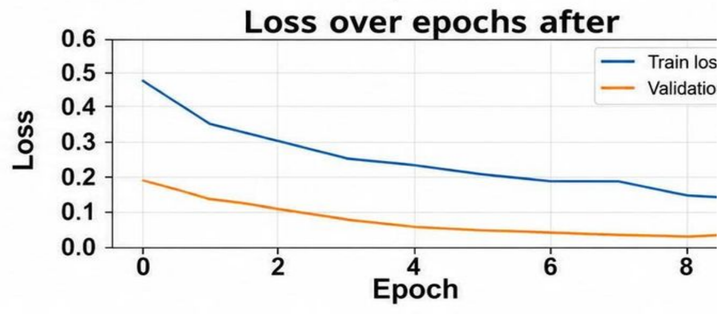
Saving model

```python
model.save("helmet_model.keras")
```



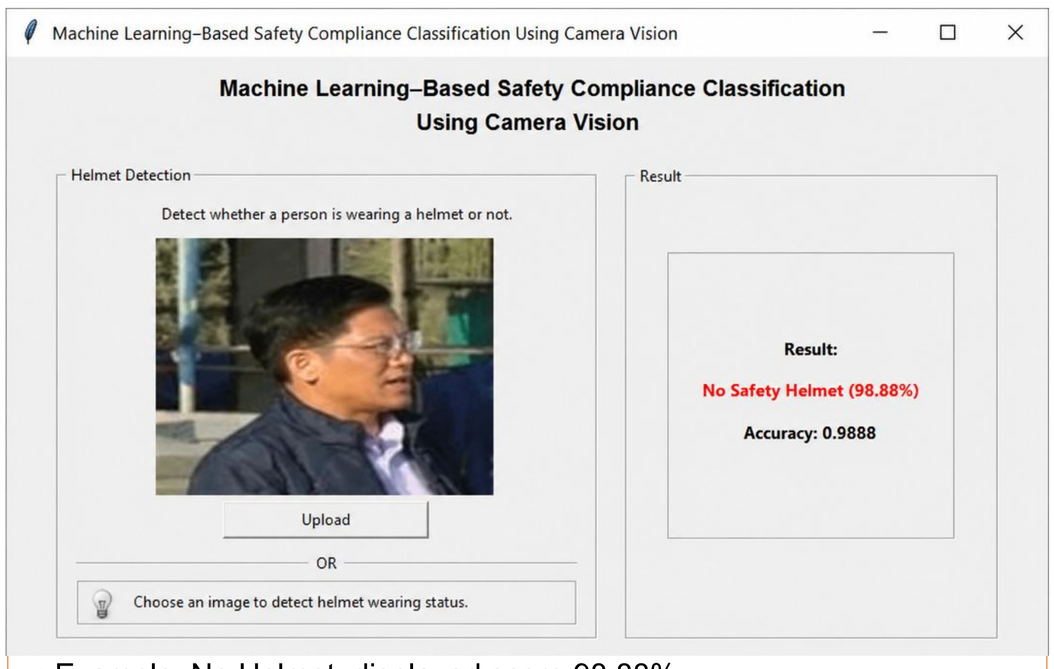

## RESULTS

### Demonstration

The GUI displays a predicted class and model score for each image. These scores are not test-set accuracy.




- Tkinter GUI for helmet-classification demonstration
- Example: Helmet; displayed score 97.63%




- Example: No Helmet; displayed score 98.88%

## CONCLUSION

The MobileNetV2 classifier was integrated into a Tkinter GUI for sample-image review. Displayed scores describe individual predictions.

Limitations:

- Binary output only; sensitive to low-light and heavy occlusion
- No persistent violation logging in current version

Future Development:

- Broader PPE detection (vests, gloves and goggles)
- Timestamped logs and a web dashboard

## Observation

- The examples show Helmet (97.63%) and No Helmet (98.88%). They are not an overall accuracy estimate.
- Displayed scores express confidence in a predicted class, not guaranteed correctness.